Lab 3 Magnetic Susceptibility

1. Calculations for Sample Tube

(1) Empty Tube

* Experimental measurements: 1.0360g, 1.0330g, 1.0344g

$$\overline{m_E} = \frac{m_{E1}+m_{E2}+m_{E3}}{3} = \frac{1.0360+1.0330+1.0344}{3}g \approx 1.0345g$$

$$S_{m_E} = \sqrt{\frac{1}{3-1}\sum_{i=1}^{3}(\overline{m_E}-m_{Ei})^2} = \sqrt{\frac{1}{2}\left[(\overline{m_E}-1.0360)^2+(\overline{m_E}-1.0330)^2+(\overline{m_E}-1.0344)^2\right]}\,g$$

$$\approx 0.0015\,g$$

95% confidence interval: $\lambda_{m_E} = 4.303 \cdot \frac{S_{m_E}}{\sqrt{3}} \approx 0.0037g$

So $m_E = (1.0345 \pm 0.0037)g$

(2) Tube with Water

* Experimental measurements: 1.8490g, 1.8509g, 1.8502g

$$\overline{m} = \frac{m_{S1}+m_{S2}+m_{S3}}{3} = \frac{1.8490+1.8509+1.8502}{3}g \approx 1.8500g$$

$$S_{m_S} = \sqrt{\frac{1}{3-1}\sum_{i=1}^{3}(\overline{m_S}-m_{Si})^2} = \sqrt{\frac{1}{2}\left[(\overline{m_S}-1.8490)^2+(\overline{m_S}-1.8509)^2+(\overline{m_S}-1.8502)^2\right]}\,g$$

$$\approx 0.0010\,g$$

95% confidence interval: $\lambda_S = 4.303 \cdot \frac{S_{m_S}}{\sqrt{3}} \approx 0.0024\,g$

So $m_S = (1.8500 \pm 0.0024)g$

(3) Mass of Water

$$\Delta m_W = m_S - m_E = (1.8500 - 1.0345)g = 0.8156g$$

$$\lambda_{\Delta m_W} = \sqrt{\lambda_{m_E}^2 + \lambda_{m_S}^2} = \sqrt{(0.0037)^2 + (0.0024)^2} \approx 0.0044g$$

$$\Delta m_W = (0.8156 \pm 0.0044)g$$

(4) Volume of Tube

Given literature density: $\rho_W = 0.997324 g/cm^3$

$$V = \frac{\Delta m_W}{\rho} = \frac{0.8156}{0.997324} cm^3 \approx 0.8178 cm^3$$

$$\lambda_V = \frac{\lambda_{\Delta m_W}}{\rho} = \frac{0.0044}{0.997324} cm^3 \approx 0.0044 cm^3$$

2. Calculation for $NiCl_2 \cdot 6H_2O$

From guideline: $\chi_m = 0.389$

$$M_{NiCl_2 \cdot 6H_2O} = 58.6934 + 2 \cdot 35.4530 + 6 \cdot 18.0153 = 237.6912 g \cdot mol^{-1}$$

(1) Calculation for Density

Experimental measurements: 2.0459g, 2.0465g, 2.0457g

$$\overline{m}_{Ni} = \frac{2.0459 + 2.0465 + 2.0457}{3} g = 2.0460g$$

$$S_{Ni} = \sqrt{\frac{1}{2} \cdot [(2.0459 - 2.0460)^2 + (2.0465 - 2.0460)^2 + (2.0457 - 2.0460)^2]} \approx 0.0004g$$

95% confidence interval: $\lambda_{Ni} = 4.3030 \cdot \frac{S_{Ni}}{\sqrt{3}} = 0.0010g$

So, $m_{Ni} = (2.0460 \pm 0.0010)g$

$$\Delta m_{Ni} = m_{Ni} - m_E = (2.0460 - 1.0345)g = 0.0039g$$

$$S_{\Delta m_{Ni}} = \sqrt{(0.0010)^2 + (0.0037)^2} \approx 0.0039\ g$$

So, $\Delta m_{Ni} = (1.0116 \pm 0.0039)g$

$$\rho_{Ni} = \frac{\Delta m_{Ni}}{V} = \frac{1.0116}{0.8178} g \cdot cm^{-3} \approx 1.2370\ g \cdot cm^{-3}$$

$$S_{\rho_{Ni}} = 1.2370 \cdot \sqrt{\left(\frac{0.0039}{1.0116}\right)^2 + \left(\frac{0.0044}{0.8178}\right)^2} \approx 0.0082\ g \cdot cm^{-3}$$

So, $\rho_{Ni} = (1.2370 \pm 0.0082)g\ cm^{-3}$

(2) Calculation for Magnetic Susceptibility & Moment

$$\chi_{w,Ni} = \frac{\chi_{Ni}}{\rho_{Ni}} = \frac{0.879 \times 10^{-5}}{1.2370} cm^3 \cdot g^{-1} \approx 7.1059 \times 10^{-6}\ cm^3 \cdot g^{-1}$$

$$S_{\chi_{w,Ni}} = \chi_{w,Ni} \cdot \frac{S_{\rho_{Ni}}}{\rho_{Ni}} = 7.1059 \times 10^{-6} \cdot \frac{0.0082}{1.2370} \approx 0.0471 \times 10^{-6}\ cm^3 \cdot g^{-1}$$

So, $\chi_{w,Ni} = (7.1059 \pm 0.0471)\ cm^3 \cdot g^{-1}$

From manual: $\chi_w = \frac{\chi_m}{M} \chi - 0.720(1-\chi) \cdot 10^{-6}\ cm^3 \cdot g^{-1}$

$$\chi_{m,Ni} = \frac{M}{\chi}\left[\chi_{w,Ni} + 0.720(1-\chi) \cdot 10^{-6}\right]$$

$$= \frac{237.9612}{0.389}\left[7.1059 + 0.720 \cdot (1 - 0.389)\right] \times 10^{-6}$$

$$\approx 4.6107 \times 10^{-3}\ cm^3 \cdot mol^{-1}$$

$$S_{\chi_{m,Ni}} = \frac{M}{\chi} S_{\chi_{w,Ni}} = \frac{237.9611}{0.389} \cdot 0.0471 \times 10^{-6} \approx 0.0288 \times 10^{-3}\ cm^3 \cdot mol^{-1}$$

So, $\chi_{m,Ni} = (4.6107 \pm 0.0288) \times 10^{-3}\ cm^3 \cdot mol^{-1}$

From literature: $\chi_{m,Ni} = 4240\times10^{-6}\,cm^3\,mol^{-1} = 4.240\times10^{-3}\,cm^3\cdot mol^{-1}$

error rate $= \frac{4.6107-4.240}{4.240}\times100\% \approx 8.7429\%$

Room temperature: $T = 23.9\,^{\circ}C = 297.05\,K$

From manual: $\mu_m = 797.7\,\mu_B\sqrt{T\chi_m}$

$$\mu_{m,Ni} = 797.7\cdot\sqrt{297.05\cdot4.240\cdot10^{-6}\cdot\chi_{m,Ni}}\,\mu_B$$

$$\approx 3.3094\,\mu_B$$

$$S_{\mu_{m,Ni}} = \pm\mu_{m,Ni}\frac{S_{\chi_{m,Ni}}}{\chi_{m,Ni}} \approx 0.0104\,\mu_B$$

So, $\mu_{m,Ni} \approx (3.3094\pm0.0104)\,\mu_B$

From manual: $\mu_s = \mu_B\sqrt{n(n+2)}$

As n is positive: $n = -1+\sqrt{1+\left(\frac{\mu_s}{\mu_B}\right)^2}$

$$n_{Ni} = -1+\sqrt{1+3.3094^2} \approx 2.4572$$

$$S_n = \frac{\mu}{\sqrt{1+\mu^2}}S_\mu \approx 0.0100$$

So, $n_{Ni} = 2.4572\pm0.0100$

3. Calculation for $FeCl_3 \cdot 6H_2O$

(1) Calculation for Density

From guideline: $\chi_m = 0.352$

Experimental measurement: 2.0279g, 2.0290g, 2.0316g

$$\overline{m_{Fe}} = \frac{2.0279+2.0290+2.0316}{3}\,g = 2.0295\,g$$

$$\overline{S_{Fe}} = \sqrt{\frac{1}{2}\cdot[(2.0295-2.0279)^2+(2.0295-2.0290)^2+(2.0295-2.0316)^2]} \approx 0.0019\,g$$

95% confidence interval: $\lambda_{m_{Fe}} = 4.303 \cdot \frac{0.0019}{\sqrt{3}} \approx 0.0047\,g$

So, $m_{Fe} = (2.0295 \pm 0.0047)\,g$

$$\Delta m_{Fe} = m_{Fe} - m_E = (2.0295 - 1.0345)\,g = 0.9950\,g$$

$$S_{\Delta m_{Fe}} = \sqrt{\lambda^2_{m_{Fe}} + \lambda^2_{m_E}} = \sqrt{(0.0047)^2+(0.0037)^2} \approx 0.0060\,g$$

So, $\Delta m_{Fe} = (0.9950 \pm 0.0060)\,g$

$$\rho_{Fe} = \frac{\Delta m_{Fe}}{V} = \frac{0.9950}{0.8178}\,g\cdot cm^{-3} \approx 1.2167\,g\cdot cm^{-3}$$

$$S_{\rho_{Fe}} = \rho_{Fe} \cdot \sqrt{\left(\frac{0.0060}{0.9950}\right)^2 + \left(\frac{0.0144}{0.8178}\right)^2} \approx 0.0099\,g\cdot cm^{-3}$$

So, $\rho_{Fe} = 1.2168 \pm 0.0099\,g\cdot cm^{-3}$

(2) Calculation for Magnetic Susceptibility & Moment

$$\chi_{w,Fe} = \frac{\chi_{Fe}}{\rho_{Fe}} = \frac{2.458\times10^{-5}}{1.2168}\,g\cdot cm^{-3} \approx 2.0201\times10^{-5}\,g\cdot cm^{-3}$$

$$S_{\chi_{w,Fe}} = \chi_{w,Fe} \cdot \frac{S_{\rho_{Fe}}}{\rho_{Fe}} \approx 2.0201\times10^{-5} \cdot \frac{0.0099}{1.2168}\,g\cdot cm^{-3} \approx 0.0164\times10^{-5}\,g\cdot cm^{-3}$$

So, $\chi_{w,Fe} = (2.0201 \pm 0.0164) \times 10^{-5}\ g \cdot cm^{-3}$

$M_{FeCl_3 \cdot 6H_2O} = 55.845 + 3 \cdot 35.453 + 6 \cdot 18.0153 = 270.2958\ g \cdot mol^{-1}$

Given $\chi_w = \frac{\chi_m}{M}\chi - 0.720(1-\chi) \cdot 10^{-6}\ cm^3 g^{-1}$

$$\chi_{m,Fe} = \frac{M}{\chi}\left[\chi_{w,Ni} + 0.720(1-\chi) \cdot 10^{-6}\right]$$

$$= \frac{270.2958}{0.352}\left[20.201 + 0.720 \cdot (1-0.352)\right] \times 10^{-6}$$

$$\approx 1.5870 \times 10^{-2}\ cm^3 \cdot mol^{-1}$$

$$S_{\chi_{m,Fe}} = \frac{M}{\chi} \cdot S_{\chi_{w,Fe}} = \frac{270.2958}{0.352} \cdot 0.0164 \times 10^{-5} \approx 0.0126 \times 10^{-2}\ cm^3 mol^{-1}$$

So, $\chi_{m,Fe} = (1.5870 \pm 0.0126) \times 10^{-2}\ cm^3 \cdot mol^{-1}$

Literature value: $\chi_{m,Fe}^{Lit} = 1.5250 \times 10^{-6}\ cm^3 \cdot mol^{-1}$

error rate $= \frac{1.5870 - 1.5250}{1.5250} \times 100\% \approx 4.0656\%$

From manual: $\mu_m = 797.7 \mu_B \sqrt{T\chi_m}$

$$\mu_{m,Fe} = 797.7 \cdot \sqrt{297.05 \cdot 4\pi \cdot 10^{-6} \cdot \chi_{m,Fe}}\ \mu_B$$

$$\approx 6.1397\ \mu_B$$

$$S_{\mu_{m,Fe}} = \frac{1}{2}\mu_{m,Fe}\frac{S_{\chi_{m,Fe}}}{\chi_{m,Fe}} = \frac{6.1397}{2} \cdot \frac{126}{15870} \approx 0.0244$$

So, $\mu_{m,Ni} \approx (6.1397 \pm 0.0244)\mu_B$

From manual: $\mu_s = \mu_B\sqrt{n(n+2)}$

As n is positive: $n = -1 + \sqrt{1 + \left(\frac{\mu_s}{\mu_B}\right)^2}$

$$n_{Fe} = -1 + \sqrt{1 + 6.1397^2} \approx 5.2206$$

$$S_n = \frac{\mu}{\sqrt{1+\mu^2}} S_\mu = \frac{5.2206}{\sqrt{1+5.2206^2}} \cdot 0.0244 \approx 0.0240$$

So, $n_{Fe} = 5.2206 \pm 0.0240$

4. Calculation for Literature Values of μ_m and n

(1) $NiCl_2 \cdot 6H_2O$

Literature: $\chi_m = 4.2400 \times 10^{-3}\ cm^3 mol^{-1}$

$$\mu_{m,lit} = 797.7\, \mu_B \sqrt{T \chi_m}$$

$$= 797.7 \cdot \sqrt{297.05 \cdot 4.2400 \times 10^{-3} \cdot 4\pi \cdot 10^{-6}}\ \mu_B$$

$$\approx 3.1735$$

$$n_{lit} = -1 + \sqrt{1 + \mu_{lit}^2} \approx 2.3274$$

$$\text{error } \mu_m = \frac{3.3094 - 3.1735}{3.1735} \times 100\% \approx 4.2823\%$$

$$\text{error } n = \frac{2.4572 - 2.3274}{2.3274} \times 100\% \approx 5.5770\%$$

(2) $FeCl_3 \cdot 6H_2O$

Literature: $\chi_m = 1.5250 \times 10^{-2}\ cm^3 mol^{-1}$

$$\mu_{m,lit} = 797.7\, \mu_B \sqrt{T \chi_m}$$

$$= 797.7 \cdot \sqrt{297.05 \cdot 1.5250 \times 10^{-2} \cdot 4\pi \cdot 10^{-6}}\ \mu_B$$

$$\approx 6.0186$$

$$n_{lit} = -1 + \sqrt{1+\mu_{lit}^2} \approx 5.1011$$

$$\text{error } \mu_m = \frac{6.1397-6.0186}{6.0186} \times 100\% \approx 2.021\%$$

$$\text{error } n = \frac{5.2206-5.1011}{5.1011} \times 100\% \approx 2.3426\ \%$$